Complex Number 17

1. a. Express $\frac{1+i\sqrt{3}}{1+i}$ in real-imaginary form.

   b. Write $1+i$ and $1+i\sqrt{3}$ in mod-arg form and hence express $\frac{1+i\sqrt{3}}{1+i}$ in mod-arg form.

   c. Hence find $\cos\frac{\pi}{12}$ in surd form.

2. Let $z = (\sqrt{3}+1) + (\sqrt{3}-1)i$

   a. By writing $\frac{\pi}{12}$ as $\frac{\pi}{3} - \frac{\pi}{4}$, show that $\tan\frac{\pi}{12} = \frac{\sqrt{3}-1}{\sqrt{3}+1}$

   b. Hence write z in mod-arg form.

3. Let $z_1 = 1 + 5i$ an$z_2 = 3 + 2i$ d, and let $z = \frac{z_1}{z_2}$

   a. Find $|z|$ without finding z

   b. Find $\tan(\tan^{-1} 5 - \tan^{-1}\frac{2}{3})$, and hence find $\arg z$ without finding z.

   c. Hence write z in the form $x + iy$, where x and y are real.

4. Show that for any non-zero complex number z:

   a. $z\bar{z} = |z|^2$

   b. $\arg(z^2) = 2\arg(z)$

   c. if $|z| = 1$ then $\bar{z} = z^{-1}$

5. The complex number z satisfies the equation $|z-1| = 1$. Square both sides and hence show that $|z|^2 = 2Re(z)$

6. If z is a complex number and $|2z-1| = |z-2|$, prove that $|z| = 1$.

7. Let $z = \operatorname{cis}\theta$ and $w = \operatorname{cis}\varphi$, that is $|z| = |w| = 1$. Evaluate $z + w$ in mod-arg form and hence show that $\arg(z+w) = \frac{1}{2}(\arg z + \arg w)$

8. Let $z = 1 + \cos\theta + i\sin\theta$.

   a. Show that $|z| = 2\cos\frac{\theta}{2}$ and $\arg z = \frac{\theta}{2}$.

   b. Hence show that $z^{-1} = \frac{1}{2} - \frac{1}{2}i\tan\frac{\theta}{2}$.

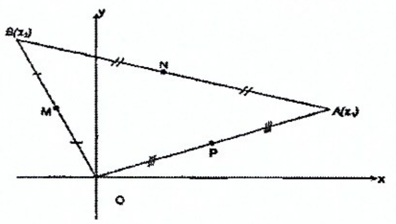
9. $|z-1| = 1$. Sketch the locus of the point P representing z on an Argand diagram. Hence deduce that $\arg(z-1) = \arg(z^2)$.

10. $z = x + iy$ is such that $\frac{z-i}{z+1}$ is purely imaginary. Find the equation of the locus of the point P representing z and show this locus on an Argand diagram.

11. $Re\left(z - \frac{1}{z}\right) = 0$. Find the equation of the locus of the point P representing z on an Argand diagram and sketch this locus.

12. If $\arg(z-2) = \arg(z+2) + \frac{\pi}{3}$, show that the locus of the point P representing z on an Argand diagram is an arc of a circle and find the centre and radius of this circle.

13. ΔABO lies on an Argand diagram. Points A and B represent the complex numbers z_1 and z_2 respectively. M, N and P are the midpoints of OB, AB and OA respectively.



   a. Which complex number is represented by N?

   b. Express $\overrightarrow{AM}$ and $\overrightarrow{BP}$ in terms of z_1 and z_2

   c. Hence simplify $\overrightarrow{ON} + \overrightarrow{AM} + \overrightarrow{BP}$

14. a. Solve $z^5 + 1 = 0$ by using De Moivre's Theorem. (Leave your answer in modulus-argument form).
   b. Explain why the solution of $z^4 - z^3 + z^2 - z + 1 = 0$ are non-real solutions of $z^5 + 1 = 0$
   c. If $z^4 - z^3 + z^2 - z + 1 = 0$ where $z = cis\theta$, show that $4\cos^2\theta - 2\cos\theta - 1 = 0$
   d. Hence find the exact value of $\cos\frac{3\pi}{5}$.

15. If $z = \cos\theta + i\sin\theta$
   a. By De Moivre's Theorem, show that $\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$.
   b. Hence show that polynomial equation $16x^5 - 20x^3 + 5x - 1 = 0$ has roots $\cos\frac{2\pi}{5}$, $\cos\frac{4\pi}{5}$, $\cos\frac{6\pi}{5}$, $\cos\frac{8\pi}{5}$ and 1.
   c. Hence deduce that $\cos\frac{2\pi}{5} \times \cos\frac{4\pi}{5} = -\frac{1}{4}$.

16. Shade in the region satisfied by both the inequalities $|z - 2 - 2i| \le |z|$ and $Re(z) \ge Im(z)$.

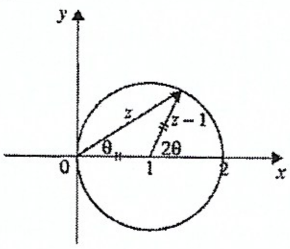
17. Let z be the complex number whose modulus is $2\sqrt{2}$ and whose argument is $\frac{3\pi}{4}$.
   a. Find z^6. Write your answer in the form $x + iy$.
   b. Find in modulus-argument form the three cube roots of z.
   c. Find the smallest integer n such that z^n is a positive real number.

18. Let $z = \cos\theta + i\sin\theta$, where $0 < \theta < \frac{\pi}{2}$ as shown in the diagram.
   a. Explain why z lies on the circle $|z| = 1$.
   b. Show that $|z + 1| = 2\cos\left(\frac{\theta}{2}\right)$.
   c. Explain geometrically why $\arg(z - 1) - \arg(z + 1) = \frac{\pi}{2}$.
   d. Hence show that $|z - 1| = 2\sin\left(\frac{\theta}{2}\right)$
   e. By finding the modulus and argument show that $\frac{z-1}{z+1} = \tan\left(\frac{\theta}{2}\right)i$.

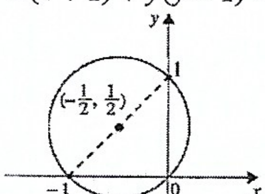
Answer

1. a. $\frac{1}{2}\left((\sqrt{3}+1) + i(\sqrt{3}-1)\right)$ b. $\sqrt{2}cis\frac{\pi}{12}$
   c. $\frac{1}{2\sqrt{2}}(\sqrt{3}+1)$
2. b. $2\sqrt{2}cis\frac{\pi}{12}$
3. a. $\sqrt{2}$ b. $\frac{\pi}{4}$ c. $1 + i$
7. $z + w = 2\cos\left(\frac{(\theta-\emptyset)}{2}\right)cis\left(\frac{(\theta+\emptyset)}{2}\right)$

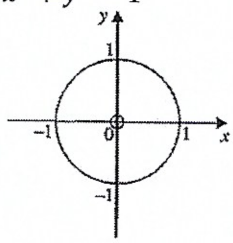
9.



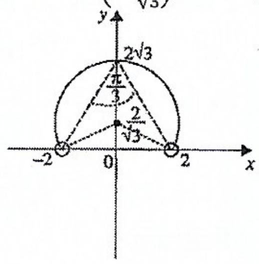
10. $x(x+1) + y(y-1) = 0$



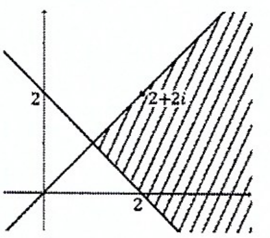
11. $x = 0, y = 0$ or $x^2 + y^2 = 1$



12. Centre $\left(0, \frac{2}{\sqrt{3}}\right)$, radius $\frac{4}{\sqrt{3}}$



13. a. $\overrightarrow{ON} = \frac{1}{2}(z_1 + z_2)$ b. $\overrightarrow{AM} = \frac{z_2}{2} - z_1$, $\overrightarrow{BP} = \frac{z_1}{2} - z_2$ c. 0
14. a. $cis\left(\frac{\pi+2k\pi}{5}\right)$, $k = 0, 1, 2, 3, 4$ c. hints: $z + \frac{1}{z} = 2\cos\theta$ d. $\frac{1-\sqrt{5}}{4}$
16.



17. a. $z^6 = 512i$ b. $\sqrt{2}cis\frac{\pi}{4}$, $\sqrt{2}cis\frac{11\pi}{12}$, $\sqrt{2}cis\frac{19\pi}{12}$ c. $n = 8$